

## Truong Vu-Quang

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### Personal Statement

Dedicated AI researcher with a strong foundation in computer science and a focus on developing cutting-edge solutions in computer vision, especially efficient object detection. Graduated top of my class, I aim to deepen my expertise in artificial intelligence, contributing to groundbreaking research and technological advancements.

### Education

#### Hanoi University of Science and Technology (HUST)

Hanoi, Vietnam

Bachelor of Computer Science, Talented Program.

2019 - 2023

- **CGPA:** 3.90/4.0. Graduated #1 in the program.
- **Thesis:** SPARTA-GEMSTONE: A two-phase approach for efficient node placement in 3D wireless sensor networks under Q-Coverage and Q-Connectivity constraints.

### Experience

#### Mohamed bin Zayed University of Artificial Intelligence (MBZUAI)

Remote

Visiting Researcher - Supervisor: Assist. Prof. Yutong Xie

11/2025 - Now

- Conducted research in Medical AI under the supervision of Assist. Prof. Yutong Xie.

#### Qualcomm AI Research

Hanoi, Vietnam

AI Resident - Supervisor: Prof. Minh Hoai Nguyen

04/2025 - 06/2025

- Continued the work at VinAI on low-rank prompt adaptation for open-vocabulary object detection, resulting in one paper accepted at ICCV 2025 Workshop.

#### VinAI Research

Hanoi, Vietnam

AI Resident - Supervisor: Prof. Minh Hoai Nguyen

02/2023 - 03/2025

- **Efficiency-preserving Scene-adaptive Object Detection:** Developed a framework for multi-camera video streams, achieving state-of-the-art precision-efficiency trade-offs (4.65 mAP improvement). Accepted for oral presentation at BMVC 2024.
- **Low-Rank Prompt Adaptation for Open-Vocabulary Object Detection:** Pioneered a framework that optimizes textual prompts, achieving state-of-the-art performance across five datasets with minimal annotation requirements.
- **Unidentified License Plate Detection:** Built a detection pipeline for obscured license plates, integrating OCR, and was awarded Best Applied Rotation Project in Q2 2024.

#### MSO Lab, HUST

Hanoi, Vietnam

Member - Supervisor: Assoc. Prof. Thi Thanh Binh Huynh

10/2020 - Present

- Conducted research on target and area coverage problems in wireless sensor networks (WSNs) and mentored new lab members. Three papers accepted in *Journal of Network and Computer Applications* (Q1, IF 7.7).

#### FPT Software

Hanoi, Vietnam

AI Engineer Intern

08/2022 - 02/2023

- Developed a chatbot using Rasa to assist the IT Helpdesk team, automating user inquiries and improving support response times.

### Publications

Conference Proceedings

1. **Truong Vu**, Minh Khoi Ho, and Yutong Xie, “Attention-Based Prototype Calibration for Multi-Rater Few-Shot Medical Image Segmentation,” in *MICCAI 2026*.  
2. Z. Zhang\*, **V. Q. Truong\***, and M. Hoai, “Low-Rank Prompt Adaptation for Open-Vocabulary Object Detection,” in *ICCV 2025 Workshop*.  
3. Z. Zhang, **V. Q. Truong**, and M. Hoai, “Efficiency-preserving scene-adaptive object detection,” in *35th British Machine Vision Conference 2024*.

Journal Articles

1. **Truong Q. Vu**, Minh T. Trinh, Hien Q. Ta, Hanh T. Nguyen, and Binh T T. Huynh, “STEAM: Securing the connections in wireless sensor networks with varying target priority,” *Computer Networks*, vol. 277, p. 112024, 2026.  
2. **Q. T. Vu**, T. M. Trinh, T. H. Nguyen, V. C. Trinh, T. T. B. Huynh, X. T. Nguyen, and C. P. Huynh, “SPARTA-GEMSTONE: A two-phase approach for efficient node placement in 3D WSNs under Q-Coverage and Q-Connectivity constraints,” *Journal of Network and Computer Applications*, vol. 239, p. 104175, 2025.  
3. **Q. T. Vu**, P. T. Nguyen, T. H. Nguyen, T. T. B. Huynh, V. C. Trinh, and M. Gidlund, “Striking the perfect balance: Multi-objective optimization for minimizing deployment cost and maximizing coverage with harmony search,” *Journal of Network and Computer Applications*, vol. 232, p. 104006, 2024.  
4. N. T. Hanh, H. T. T. Binh, **V. Q. Truong**, N. P. Tan, and H. C. Phap, “Node placement optimization under q-coverage and q-connectivity constraints in wireless sensor networks,” *Journal of Network and Computer Applications*, vol. 212, p. 103578, 2023.

Awards and Achievements

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|----------------------------------------------------------------------------------------------|------------|
| <b>Best Applied Rotation Project</b> of Quarter 2, VinAI Research                            | 2024       |
| <b>Third prize</b> , AI Hackathon: Face Analysis Challenge, Pixta Vietnam                    | 2024       |
| 2 times <b>Sumitomo Sponsor Scholarship</b> , HUST                                           | 2023, 2020 |
| <b>Best Presentation Award</b> for the graduation thesis in the Optimization committee, HUST | 2023       |
| <b>First prize</b> , SoICT Hackathon, Tiki Track                                             | 2023       |
| <b>Consolation Prize</b> , SoICT Student Research Conference                                 | 2022       |
| 3 times <b>Study Encouragement Scholarship</b> , HUST                                        | 2022, 2021 |
| <b>Top 6 in National High School Exam</b> with Math, Physics, and English combination        | 2019       |

Certifications

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| <b>IELTS Academic:</b> 8.0 (Reading 9.0, Listening 8.5, Writing 7.0, Speaking 6.5), British Council | 2024 |
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Skills

**Technical:** Java, Python, C/C++, SQL, LaTeX, MySQL, PostgreSQL, SQLite  
**Language:** Proficient in English (reading, writing, and speaking)  
**AI Expertise:** Computer Vision  
**Others:** Academic research, teamwork, presentation skills

References

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|---------------------------------------------------|---------------------------------------------------------|
| 1. <b>Minh Hoai Nguyen</b> , <i>Professor</i>     | The University of Adelaide, Australia                   |
| <b>Email:</b> minhhoai.nguyen@adelaide.edu.au     |                                                         |
| 2. <b>Yutong Xie</b> , <i>Assistant Professor</i> | Mohamed bin Zayed University of Artificial Intelligence |
| <b>Email:</b> yutong.xie@mbzuai.ac.ae             |                                                         |
| 3. <b>Thi Thanh Binh Huynh</b> , <i>Professor</i> | Hanoi University of Science and Technology, Vietnam     |
| <b>Email:</b> binhht@soict.hust.edu.vn            |                                                         |

\*Equal contribution